

Computing and Interpreting Centre and Spread

Mean, median and mode; range and interquartile range; and choosing the pair that describes a data set honestly

Directions:

- **Worked pattern.** For 6, 2, 4, 8: the mean is $(6 + 2 + 4 + 8) \div 4 = 5$; in order the values are 2, 4, 6, 8, so the median is $(4 + 6) \div 2 = 5$ and the range is $8 - 2 = 6$.
- **Always order the values first** before finding a median, a quartile or a range. Most lost marks on this page start there.
- **Quartiles:** split the ordered values into a lower half and an upper half (leave a middle value out if there is one), then take the median of each half. Interquartile range = upper – lower.
- Show the ordered list in your work space, not just the answer.

1. Find the **mean** of 4, 7, 5, 9, 5.

Answer: _____

2. Find the **range** of 11, 17, 9, 20, 14.

Answer: _____

3. Find the **median** of 2, 9, 4, 11, 6, 8.

Answer: _____

4. Find the **mode** of 3, 5, 3, 7, 3, 5, 8.

Answer: _____

5. Eight values, already in order:

2	4	8	10	14	16	20	22
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(a) the lower quartile: _____

(b) the upper quartile: _____

(c) the interquartile range: _____

6. Five bicycles in a shop are priced, in dollars, at

12	14	13	15	46
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- (a) the mean price: _____
- (b) the median price: _____
- (c) Which of the two represents these prices better? Say why in the work space.

7. Twelve values, already in order:

1	3	5	7	9	11	13	15	17	19	21	23
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- (a) the range: _____
- (b) the interquartile range: _____

8. Five numbers have a mean of 12. Four of them are 8, 10, 15, 11. Find the fifth number.

Answer: _____

9. A bus takes two different routes to school. Its journey times, in minutes, were:

Route A	10	12	14	16	18
Route B	2	8	14	20	26

Both routes have a median time of 14 minutes.

- (a) the interquartile range for Route A: _____
- (b) the interquartile range for Route B: _____
- (c) Which route is more predictable? Back your answer with the two interquartile ranges, in the work space.

10. Five deliveries weighed, in kilograms:

12	16	20	24	28
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(a) the mean mass: _____

(b) the **mean absolute deviation** — the mean of the distances from each value to the mean mass: _____

11. One group of 4 students had a mean mark of 80. A second group of 6 students had a mean mark of 90. Find the mean mark of the two groups combined.

Answer: _____

12. Nine donations to a school fund, in dollars:

2	4	4	6	6	6	8	10	44
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(a) the mean: _____

(b) the median: _____

(c) the range: _____

(d) the interquartile range: _____

(e) A newsletter will print one measure of centre and one of spread. Which pair describes these donations honestly? Explain in the work space.